

Habit Forming

- No

How to Stop

- Taper not necessary
- Discontinuation may lead to notable deterioration in memory and behavior, which may not be restored when drug is restarted or another cholinesterase inhibitor is initiated

Pharmacokinetics

- Terminal elimination half-life approximately 7 hours
- Metabolized by CYP450 2D6 and 3A4



Drug Interactions

- Galantamine may increase the effects of anesthetics and should be discontinued prior to surgery
- Inhibitors of CYP450 2D6 and CYP450 3A4 may inhibit galantamine metabolism and raise galantamine plasma levels
- Galantamine may interact with anticholinergic agents and the combination may decrease the efficacy of both
- Cimetidine may increase bioavailability of galantamine
- May have synergistic effect if administered with cholinomimetics (e.g., bethanechol)

- Bradycardia may occur if combined with beta blockers
- Theoretically, could reduce the efficacy of levodopa in Parkinson's disease
- Not rational to combine with another cholinesterase inhibitor

Other Warnings/Precautions

- May exacerbate asthma or other pulmonary disease
- Increased gastric acid secretion may increase the risk of ulcers
- Bradycardia or heart block may occur in patients with or without cardiac impairment

Do Not Use

- If there is a proven allergy to galantamine

Special Populations

Renal Impairment

- Should be used with caution
- Not recommended for use in patients with severe renal impairment

Hepatic Impairment

- Should be used with caution

- Reduction of clearance may increase with the degree of hepatic impairment
- Not recommended for use in patients with severe hepatic impairment

Cardiac Impairment

- Should be used with caution
- Syncopal episodes have been reported with the use of galantamine

Elderly

- Clearance is reduced in elderly patients
- Use of cholinesterase inhibitors may be associated with increased rates of syncope, bradycardia, pacemaker insertion, and hip fracture in older adults with dementia



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR

or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Controlled studies have not been conducted in pregnant women
- Animal studies do not show adverse effects
- * Not recommended for use in pregnant women or in women of childbearing potential

Breast Feeding

- Unknown if galantamine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Galantamine is not recommended for use in nursing women

The Art Of Psychopharmacology

Potential Advantages

- Alzheimer disease with cerebrovascular disease
- Theoretically, nicotinic modulation may provide added therapeutic benefits for memory and behavior in some Alzheimer patients
- Theoretically, nicotinic modulation may also provide efficacy for cognitive disorders other than Alzheimer disease

Potential Disadvantages

- Patients who have difficulty taking a medication twice daily

Primary Target Symptoms

- Memory loss in Alzheimer disease
- Behavioral symptoms in Alzheimer disease
- Memory loss in other dementias

Pearls

- Dramatic reversal of symptoms of Alzheimer disease is not generally seen with cholinesterase inhibitors
- Can lead to therapeutic nihilism among prescribers and lack of an appropriate trial of a cholinesterase inhibitor
- * Perhaps only 50% of Alzheimer patients are diagnosed, and only 50% of those diagnosed are treated, and only 50% of those treated are given a cholinesterase inhibitor, and then only for 200 days in a disease that lasts 7–10 years
- Must evaluate lack of efficacy and loss of efficacy over months, not weeks
- * Treats behavioral and psychological symptoms of Alzheimer dementia as well as cognitive symptoms (i.e., especially apathy, disinhibition, delusions, anxiety, cooperation, pacing)
- Patients who complain themselves of memory problems may have depression, whereas patients whose spouses or children complain of the patient's memory problems may have Alzheimer disease

- Treat the patient but ask the caregiver about efficacy
- What you see may depend upon how early you treat
- The first symptoms of Alzheimer disease are generally mood changes; thus, Alzheimer disease may initially be diagnosed as depression
- Women may experience cognitive symptoms in perimenopause as a result of hormonal changes that are not a sign of dementia or Alzheimer disease
- Aggressively treat concomitant symptoms with augmentation (e.g., atypical antipsychotics for agitation, antidepressants for depression)
- If treatment with antidepressants fails to improve apathy and depressed mood in the elderly, it is possible that this represents early Alzheimer disease and a cholinesterase inhibitor like galantamine may be helpful
- What to expect from a cholinesterase inhibitor:
 - Patients do not generally improve dramatically although this can be observed in a significant minority of patients
 - Onset of behavioral problems and nursing home placement can be delayed
 - Functional outcomes, including activities of daily living, can be preserved
 - Caregiver burden and stress can be reduced

- Delay in progression in Alzheimer disease is not evidence of disease-modifying actions of cholinesterase inhibition
- Cholinesterase inhibitors like galantamine depend upon the presence of intact targets for acetylcholine for maximum effectiveness and thus may be most effective in the early stages of Alzheimer disease
- The most prominent side effects of galantamine are gastrointestinal effects, which are usually mild and transient
- For patients with intolerable side effects, generally allow a washout period with resolution of side effects prior to switching to another cholinesterase inhibitor
- Weight loss can be a problem in Alzheimer patients with debilitation and muscle wasting
- Women over 85, particularly with low body weights, may experience more adverse effects
- Use with caution in underweight or frail patients
- Cognitive improvement may be linked to substantial (>65%) inhibition of acetylcholinesterase
- * Galantamine is a natural product present in daffodils and snowdrops
- New extended-release formulation allows for once daily dosing
- * Novel dual action uniquely combines acetylcholinesterase inhibition with allosteric nicotine modulation
- * Novel dual action should theoretically enhance cholinergic actions but incremental clinical benefits have been difficult to demonstrate

* Actions at nicotinic receptors enhance not only the release of acetylcholine but also that of other neurotransmitters, which may boost attention and improve behaviors caused by deficiencies in those neurotransmitters in Alzheimer disease

- Some Alzheimer patients who fail to respond to another cholinesterase inhibitor may respond when switched to galantamine
- Some Alzheimer patients who fail to respond to galantamine may respond to another cholinesterase inhibitor
- To prevent potential clinical deterioration, generally switch from long-term treatment with one cholinesterase inhibitor to another without a washout period

* Galantamine may slow the progression of mild cognitive impairment to Alzheimer disease

* May be useful for dementia with Lewy bodies (DLB, constituted by early loss of attentiveness and visual perception with possible hallucinations, Parkinson-like movement problems, fluctuating cognition such as daytime drowsiness and lethargy, staring into space for long periods, episodes of disorganized speech)

- May decrease delusions, apathy, agitation, and hallucinations in dementia with Lewy bodies

* May be useful for vascular dementia (e.g., acute onset with slow stepwise progression that has plateaus, often with gait abnormalities, focal signs, imbalance, and urinary incontinence)

- May be helpful for dementia in Down's syndrome

- Suggestions of utility in some cases of treatment-resistant bipolar disorder
- Theoretically, may be useful for ADHD, but not yet proven
- Theoretically, could be useful in any memory condition characterized by cholinergic deficiency (e.g., some cases of brain injury, cancer chemotherapy-induced cognitive changes, etc.)

Suggested Reading

Bentue-Ferrer D , Tribut O , Polard E , Allain H . Clinically significant drug interactions with cholinesterase inhibitors: a guide for neurologists . *CNS Drugs* 2003 ;**17** :947–63 .

Bonner LT , Peskind ER . Pharmacologic treatments of dementia . *Med Clin North Am* 2002 ;**86** :657–74 .

Coyle J , Kershaw P . Galantamine, a cholinesterase inhibitor that allosterically modulates nicotinic receptors: effects on the course of Alzheimer's disease . *Biol Psychiatry* 2001 ;**49** :289–99 .

Jones RW . Have cholinergic therapies reached their clinical boundary in Alzheimer's disease ? *Int J Geriatr Psychiatry* 2003 ;**18** (Suppl 1):S7 –13.

Olin J , Schneider L . Galantamine for Alzheimer's disease . *Cochrane Database Syst Rev* 2002 ;(3):CD001747 .

.....
Stahl SM . Cholinesterase inhibitors for Alzheimer's disease . *Hosp Pract (Off Ed)* 1998 ;**33** :131 –6 .
.....

Stahl SM . The new cholinesterase inhibitors for Alzheimer's disease, part 1 . *J Clin Psychiatry* 2000 ;**61** :710–11 .
.....

Stahl SM . The new cholinesterase inhibitors for Alzheimer's disease, part 2 . *J Clin Psychiatry* 2000 ;**61** :813–14 .

Guanfacine

Therapeutics

Brands

- Intuniv
- Tenex
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: norepinephrine receptor agonist (N-RA)
- Centrally acting alpha 2A agonist; antihypertensive; nonstimulant for ADHD

Commonly Prescribed for

(bold for FDA approved)

- **Hypertension**
- **Attention deficit hyperactivity disorder (ADHD) in children ages 6–17 (Intuniv, adjunct and monotherapy)**

- Oppositional defiant disorder
- Conduct disorder
- Pervasive developmental disorders
- Motor tics
- Tourette's syndrome

How the Drug Works

- For ADHD, theoretically has central actions on postsynaptic alpha 2A receptors in the prefrontal cortex
- Guanfacine is 15–20 times more selective for alpha 2A receptors than for alpha 2B or alpha 2C receptors
- The prefrontal cortex is thought to be responsible for modulation of working memory, attention, impulse, control, and planning
- For hypertension, stimulates alpha 2A adrenergic receptors in the brain stem, reducing sympathetic outflow from the CNS and decreasing peripheral resistance, renal vascular resistance, heart rate, and blood pressure

How Long Until It Works

- For ADHD, can take a few weeks to see maximum therapeutic benefits
- Blood pressure may be lowered 30–60 minutes after first dose; greatest reduction seen after 2–4 hours

- May take several weeks to control blood pressure adequately

If It Works

- The goal of treatment of ADHD is reduction of symptoms of inattentiveness, motor hyperactivity, and/or impulsiveness that disrupt social, school, and/or occupational functioning
- Continue treatment until all symptoms are under control or improvement is stable and then continue treatment indefinitely as long as improvement persists
- Some studies of up to 2 years
- Reevaluate the need for treatment periodically
- Treatment for ADHD begun in childhood may need to be continued into adolescence and adulthood if continued benefit is documented

If It Doesn't Work

- Consider adjusting dose or switching to another agent
- Consider behavioral therapy
- Consider the presence of noncompliance and counsel patient and parents
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., bipolar disorder, substance abuse, medical illness, etc.)

Best Augmenting Combos for Partial Response or Treatment Resistance

- Best to attempt another monotherapy prior to augmenting for ADHD
- Possibly combination with stimulants (with caution)
- Combinations for ADHD should be for the expert, while monitoring the patient closely, and when other treatment options have failed
- Chlorthalidone, thiazide-type diuretics, and furosemide for hypertension

Tests

- Blood pressure should be checked regularly during treatment

Side Effects

How Drug Causes Side Effects

- Excessive actions on alpha 2A receptors, nonselective actions on alpha 2B and alpha 2C receptors

Notable Side Effects

- Sedation, dizziness
- Dry mouth, constipation, abdominal pain
- Fatigue, weakness

- Hypotension



Life-Threatening or Dangerous Side Effects

- Sinus bradycardia, hypotension (dose-related)



Weight Gain

- Reported but not expected



Sedation

- Many experience and/or can be significant in amount
- Some patients may not tolerate it
- Can abate with time
- May be less sedation with extended-release formulation

What to Do About Side Effects

- Wait
- Adjust dose
- If side effects persist, discontinue use

Best Augmenting Agents for Side Effects

- Dose reduction or switching to another agent may be more effective since most side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- Immediate-release: 1–2 mg/day
- Extended-release: 1–4 mg/day

Dosage Forms

- Immediate-release tablet 1 mg, 2 mg, 3 mg
- Extended-release: 1 mg, 2 mg, 3 mg, 4 mg

How to Dose

- Immediate-release: initial 1 mg/day at bedtime; after 3–4 weeks can increase to 2 mg/day
- Extended-release: initial 1 mg/day; can increase by 1 mg/week; maximum dose 4 mg/day



Dosing Tips

- Adverse effects are dose-related and usually transient
- Doses greater than 2 mg/day are associated with increased side effects
- If guanfacine is terminated abruptly, rebound hypertension may occur within 2–4 days
- For hypertension, dose can be raised to 2 mg/day if 1 mg/day is ineffective, but 2 mg may have no more efficacy than 1 mg

- For extended-release formulation, do not administer with high-fat meals because this increases exposure
- Extended-release tablets should not be crushed, chewed, or broken
- Extended-release and immediate-release tablets have different pharmacokinetic properties, so do not substitute on a mg-per-mg basis
- Consider dosing extended-release on a mg/kg basis (0.05 mg/kg to 0.12 mg/kg)

Overdose

- Drowsiness, lethargy, bradycardia, hypotension

Long-Term Use

- Shown to be safe and effective for treatment of hypertension
- Studies of up to 2 years in ADHD

Habit Forming

- No

How to Stop

- Taper to avoid rebound effects (nervousness, increased blood pressure)

Pharmacokinetics

- Pharmacokinetic properties differ for immediate- and extended-release formulations
- Metabolized by CYP450 3A4

Drug Interactions

- CYP450 3A inhibitors, such as nefazodone, fluoxetine, fluvoxamine, and ketoconazole, may decrease clearance of guanfacine and raise guanfacine levels significantly
- CYP450 3A inducers may increase clearance of guanfacine and lower guanfacine levels significantly
- Do not administer extended-release with high-fat meals, because this increases exposure
- Combined use with valproate may increase plasma concentrations of valproate
- Increased depressive effects when taken with other CNS depressants
- Phenobarbital and phenytoin may reduce plasma concentrations of guanfacine

Other Warnings/Precautions

- Excessive heat (e.g., saunas) may exacerbate some of the side effects, such as dizziness and drowsiness
- Titrate slowly and monitor vital signs frequently in patients at risk for hypotension, heart block, bradycardia, syncope, cardiovascular

disease, vascular disease, cerebrovascular disease, or chronic renal failure

Do Not Use

- If there is a proven allergy to guanfacine

Special Populations

Renal Impairment

- Patients should receive lower doses

Hepatic Impairment

- Use with caution

Cardiac Impairment

- Use with caution in patients with recent myocardial infarction, severe coronary insufficiency, cerebrovascular disease
- Use with caution in patients at risk for hypotension, bradycardia, heart block, or syncope

Elderly

- Elimination half-life may be longer in elderly patients
- Elderly patients may be more sensitive to sedative effects

Children and Adolescents

- Safety and efficacy not established in children under age 6
- Some reports of mania and aggressive behavior in ADHD patients taking guanfacine

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Animal studies do not show adverse effects
- Use in women of childbearing potential requires weighing potential benefits to the mother against potential risks to the fetus

Breast Feeding

- Unknown if guanfacine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Recommended either to discontinue drug or bottle feed

The Art Of Psychopharmacology

Potential Advantages

- No known abuse potential; not a controlled substance
- Not a stimulant
- For oppositional behavior associated with ADHD
- Less sedation than clonidine

Potential Disadvantages

- Not well studied in adults with ADHD

Primary Target Symptoms

- Concentration
- Motor hyperactivity
- Oppositional and impulsive behavior
- High blood pressure



Pearls

- The extended-release formulation often is much more tolerable than the immediate-release formulation, especially for patients sensitive to peak dose sedation of the immediate-release formulation
- Guanfacine has been shown to be effective in both children and adults, and guanfacine extended-release is approved for ADHD in children ages 6–17

- Guanfacine can also be used to treat tic disorders, including Tourette's syndrome
- Although both guanfacine and clonidine are alpha 2 adrenergic agonists, guanfacine is relatively selective for alpha 2A receptors, whereas clonidine binds not only alpha 2A, 2B, and 2C receptors but also imidazoline receptors, causing more sedation, hypotension, and side effects than guanfacine
- May be used as monotherapy or in combination with stimulants for the treatment of oppositional behavior in children with or without ADHD

Suggested Reading

Arnsten AF , Scahill L , Findling RL . alpha2-Adrenergic receptor agonists for the treatment of attention-deficit/hyperactivity disorder: emerging concepts from new data . *J Child Adolesc Psychopharmacol* 2007 ;**17** (4):393 –406 .

.....

Biederman J , Melmed RD , Patel A , et al. Long-term, open-label extension study of guanfacine extended-release in children and adolescents with ADHD . *CNS Spectr* 2008 ;**13** (12):1047 –55.

.....

Posey DJ , McDougal CJ . Guanfacine and guanfacine extended-release: treatment for ADHD and related disorders . *CNS Drug Rev* 2007 ;**13** (4):465–74 .

.....

Sallee FR , Lyne A , Wigal T , McGough J . Long-term safety and efficacy of guanfacine extended-release in children and adolescents with attention-deficit/hyperactivity disorder . *J Child Adolesc Psychopharmacol* 2009 ;**19** (3):215–26 .

.....

Sallee FR , McGough J , Wigal T , et al. Guanfacine extended-release in children and adolescents with attention-deficit/hyperactivity disorder: a placebo-controlled trial . *J Am Acad Child Adolesc Psychiatry* 2009 ;**48** (2):155–65 .

.....

Spencer TJ , Greenbaum M , Ginsberg LD , Murphy WR . Safety and effectiveness of coadministration of guanfacine extended-release and psychostimulants in children and adolescents with attention-deficit/hyperactivity disorder . *J Child Adolesc Psychopharmacol* 2009 ;**19** (5):501–10 .

Haloperidol

Therapeutics

Brands

- Haldol
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine receptor antagonist (D-RAn)
- Conventional antipsychotic (neuroleptic, butyrophenone, dopamine 2 antagonist)

Commonly Prescribed for

(bold for FDA approved)

- **Manifestations of psychotic disorders (oral, immediate-release injection)**
- **Tics and vocal utterances in Tourette's syndrome (oral, immediate-release injection)**

- **Second-line treatment of severe behavior problems in children of combative, explosive hyperexcitability (oral)**
- **Second-line short-term treatment of hyperactive children (oral)**
- **Treatment of schizophrenic patients who require prolonged parenteral antipsychotic therapy (depot intramuscular decanoate)**
- Bipolar disorder
- Behavioral disturbances in dementias
- Delirium (with lorazepam)

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis and possibly combative, explosive, and hyperactive behaviors
- Blocks dopamine 2 receptors in the nigrostriatal pathway, improving tics and other symptoms in Tourette's syndrome

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them